

Poster: An Efficient Permissioned Blockchain with Provable Reputation Mechanism

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Abstract—Permissioned blockchains take more reliability on participants than permissionless ones. In this poster, we focus on a hierarchical scenario of permissioned blockchains, which includes three types of participants: providers, collectors, and governors. Such a scenario has many applications in the field of IoT data collection, horizontal strategic alliances, etc. Our object is to reduce the cost of the governor's transaction verification. For this purpose, we propose a reputation protocol to help the governor measure the reliability of collectors. Based on the measurement of collectors' reputations, governors can pack high-quality transactions from reliable collectors into blocks, and thus the cost of verifying transactions can be decreased effectively. Through theoretical analysis, our protocol dramatically reduces the verification loss of governors.

Index Terms—Permissioned blockchain, Transaction verification, Reputation mechanism, Hierarchical structure

I. INTRODUCTION AND SYSTEM MODEL

In recent years, the popularity of permissioned blockchains has increased visibly, much due to the availability of designed permissions to different users on the network. Compared with permissionless blockchains, permissioned blockchains are relatively closed and only allow authorized participants to access and play designated roles. Such permissioned requirements provide an extra security layer to unleash the power of accountability and reliability in blockchains for the regulated institutes, such as governments and private companies.

A key benefit of adopting a permissioned system is to tie a participant's trustfulness in the current task to his long-term behalf, which helps to build a reputation protocol [2] in a permissioned system much more easily. As the name suggests, one's reputation is evaluated according to his historical behaviors and can work as a measurement of his reliability. There are two obvious advantages of applying the reputation mechanism into blockchain systems. One is that blockchain's immutability and decentralization are just in line with the reputation system. The other is that reputation can serve as an incentive to different blockchain applications.

In this poster, we study a permissioned blockchain system with a hierarchical structure, in which there are three kinds of nodes: *providers*, *collectors* and *governors*. Such a permissioned blockchain system resembles Hyperledger Fabric in which the identities of participants are all maintained by an

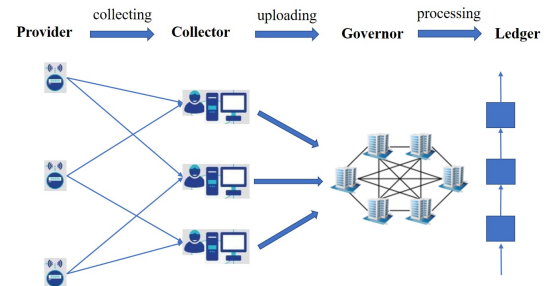


Fig. 1. Hierarchical model of the network

Identity Manager (IM). Providers offer original transactions to collectors. Collectors verify the transactions, label them as +1 or -1, and then send the labeled transactions to governors. +1 indicates the transaction is considered *valid*, while -1 otherwise. Governors are responsible for validating transactions, proposing blocks, and maintaining the ledger.

For different purposes, collectors may provide inconsistent labels for the same transaction. How to make a judgment on the labeled transaction according to opinions from several collectors is a key issue for governors. The main contributions are as follows. First, we build a reputation mechanism to measure the reliability of collectors based on their historical behaviors. Second, we propose a verification strategy that helps the governors selectively verify part of the received transactions to avoid the invalid ones as far as possible.

II. REPUTATION PROTOCOL

We suppose each provider p_i is connected with u_i collectors $C_i = \{c_i(1), c_i(2), \dots, c_i(u_i)\}$ in the hierarchical structure. Under our reputation protocol, for each provider p_i , governors would maintain a local reputation list $\vec{R}_i = (r_i(1), r_i(2), \dots, r_i(u_i))$ on collectors, where $r_i(j)$ reflects the reliability of the j -th collector regarding provider p_i . Initially $\vec{R}_i$ is a zero vector.

Our reputation protocol runs in multi-rounds, each containing three phrases, shown in Fig. 1.

- *Collecting*. The provider offers an original transaction tx to

several collectors.

- **Uploading.** Collectors verify the received transactions and label them. These labeled transactions tx are then uploaded to governors.
- **Processing.** In each round, a leader is elected by PoS protocol from all governors. An original transaction tx could be contained in several labeled transactions sent by collectors. The leader chooses a collector $c_i(k) \in C_i$ with the probability proportional to $\exp(\eta r_i(k))$, and decides whether to verify tx according to its label by this collector. If the transaction is labeled as -1 , the governor does not check it and directly append it to *UncheckedList*. If it has a label of $+1$, then the governor would verify it and append it to *TXList* once tx is valid, or to *InvalidList* otherwise. Meanwhile, the leader updates the reputations of all collectors according to the verification result, which will be introduced in the following. At the end of one round, the leader proposes a block, and governors reach consensus on the block.

Reputation Updating: For each transaction that has been validated by the leader, the reputation of collectors that label this transaction correctly will keep the same. In the meantime, reputations of those who report the opposite labels will decrease by one unit, i.e. $r_i(j) \leftarrow r_i(j) - 1$. Transactions that are not verified won't influence reputations of collectors.

Block Proposing: At the end of each round, the leader generates a block B , which contains all valid transactions in *TXList*. Formally, a block B is represented as

$$B = (s, g_{leader}, TXList = \{tx_1, \dots, tx_b\}, MT, h),$$

where s is the serial number of block B , g_{leader} is the leader in current round, *TXList* is the list of verified valid transactions, *MT* is the Merkle Tree root of (*InvalidList*, *UncheckedList*), and h is the hash value of previous block with serial number $s - 1$. After generating a new block, the leader broadcasts it to all other participants. The details of (*InvalidList*, *UncheckedList*) will be also broadcast. After receiving this information, the provider can figure out whether his valid transactions are checked or not, and then decide whether to send it again.

III. PERFORMANCE ANALYSIS

Analysis of Consensus Reaching:

- **Agreement:** For any two blocks B and B' retrieved with the same serial number s , $B = B'$;
- **Chain Integrity:** For any two blocks B and B' , if B is retrieved with serial number s while B' is with serial number $s + 1$, then the hash value h' contained in B' satisfies $h' = H(B)$, where H is a public collision-resistant hash function (CRHF);
- **No Skipping:** Once a block with the serial number s is retrieved by a node, then blocks with all previous serial numbers (i.e., $1, \dots, s - 1$) are already retrieved before;
- **Almost No Creation:** When an *honest* node perceives tx when retrieving a block B , then excluding negligible probability of λ , tx has been broadcast both by providers and collectors previously. Here λ is the security parameter of the protocol.

Efficiency Analysis of the Reputation Mechanism:

In many application scenarios, collectors are relatively close to providers such that it is easy for them to verify the transactions. By comparison, the cost of verifying a transaction by governors is relatively high. So verifying invalid transactions is just a waste of the computation of governors. For the case containing one-governor and one-provider, which can be easily extended to a general case, we define the *verification loss* as the total number of invalid transactions provided by p and further checked by governors. We analyze the efficiency of our protocol and obtain the following theorem.

Theorem 1 Consider a group of collectors who are connected with provider p . Suppose provider p offers T transactions. Let $\mathcal{L}_T$ be the accumulated expected verification loss on these transactions for the governor, and $\mathcal{S}_T^{\min}$ be the accumulated verification loss for the best-behaving collector on these transactions, with any parameter $\eta > 0$, we have

$$\mathcal{L}_T - \mathcal{S}_T^{\min} \leq \frac{\ln u}{\eta} + \frac{\eta T}{2},$$

where u is the number of collectors linked to provider p .

For any T , if setting $\eta = \sqrt{\frac{\ln u}{T}}$, then we can get $\mathcal{L}_T - \mathcal{S}_T^{\min} \leq \frac{3}{2}\sqrt{T \ln u}$. We can double the value of T and reset η every time. As a result, after $T_{total} = T + 2T + \dots + 2^{\ell-1}T = (2^\ell - 1)T$ transactions, we have $\mathcal{L}_T - \mathcal{S}_T^{\min} \leq O(\sqrt{T_{total}})$, showing our protocol can greatly reduce the verification loss of governors, which is proved to be asymptotically $O(\sqrt{T_{total}})$.

IV. APPLICATION

Our protocol provides a solution to reduce network overhead for IoT systems, where the devices, edge nodes and cloud nodes can be viewed as providers, collectors and governors, respectively. Cloud nodes are trustful and maintain a blockchain to record the history. Far-end cloud nodes need to collect that meets certain conditions data from front-end devices, but the cost of transporting all data is too high. So edge nodes are introduced to be responsible for collecting and forwarding data, and then cloud nodes build a reputation protocol on edge nodes to help filter the data from the edge nodes.

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